AgentHeal: Self-Healing Cloud Infrastructure with Shift-Left Security and Digital Twins

Aadi Haldar, Raghuram Sekar, Aaditya Paul, Shravan Rajesh Menon
Department of Artificial Intelligence & Engineering
Amrita Vishwa Vidyapeetham, Coimbatore, India

***Abstract*—In modern cloud setups, microservices tend to break when we least expect them to. If an app locks up because of a deadlock, the standard Kubernetes autoscaler just adds more and more pods. That does not help. In fact, it only makes the database crash that much faster. In our tests, we found that bad code can easily slip right past reviewers during daily pull request reviews. We built AgentHeal to stop this. What we did was connect pre-merge code checks directly to live cluster self-healing in one single loop. When a developer opens a pull request, our GitHub bot looks over the code for ten common OWASP security flaws, and it checks all third-party libraries for known CVEs as well. If the pull request scores below our safety bar, we block the merge right away. From there, an agent powered by Gemini 2.0 Flash writes out a clean fix branch with unit tests that pass. Out in the running cluster, an in-memory Isolation Forest model keeps an eye on Prometheus metrics. In our tests, it caught anomalies in just about two milliseconds. KernelSHAP then tells us which metric caused the trouble. Two agents—a Q-learning policy and an LLM—work together to pick the best fix. We never act blindly. Before anything touches the cluster, we test the proposed action inside a SimPy $M/M/c$ queue simulation. If the queue would overload, we drop the action. This keeps the cluster safe. In our tests across sixty failure runs on Google Online Boutique, the system brought services back up in 2.98 seconds on average. The results were clear. Compared to default Kubernetes autoscaling, that cut down recovery time by 98%. In our tests on a live GitHub repository, it also patched all fourteen injected security flaws without needing any human help.**

***Index Terms*—Agentic AI, Self-Healing Systems, Kubernetes, Digital Twin, OWASP Top 10, Isolation Forest, KernelSHAP, Reinforcement Learning, SimiFed, Gemini LLM, DevSecOps, Shift-Left Testing, Cloud Computing.**

I. Introduction

Cloud native applications are notoriously tricky to keep healthy. In complex microservice architectures, things break in unexpected ways, and most outages ultimately trace back to two separate failures in modern engineering practices. First, fragile code routinely makes it into production. DORA's 2024 report revealed that over 60% of unplanned downtime stems directly from preventable code defects [1]. Second, once deployed, standard cluster autoscalers operate as blunt instruments. When an incident occurs, human site reliability engineers often take upwards of 45 minutes to diagnose and resolve the issue. Meanwhile, the native Kubernetes Horizontal Pod Autoscaler (HPA) only monitors coarse metrics like CPU and memory utilization. If a microservice hangs due to an internal deadlock or thread starvation, scaling up the replica count merely wastes compute budget and overburdens upstream databases [1], [9]. It does nothing to cure the underlying software disease.

Today, development and operations remain stubbornly disconnected. Shift-Left linters like Bandit [6] and SonarQube [7] check source files during continuous integration. But they operate completely blind to live cluster health, nor can they author tested patches to fix what they flag. On the operations side, monitoring agents like Prometheus and Chaos Mesh [24], [23] alert engineers after a crash has already happened, but they offer no root-cause attribution and cannot verify whether an automated fix might destabilize downstream dependencies. While recent literature explores standalone digital twins [12] or language-model program repair [15], no existing framework connects both boundaries into a single closed loop.

We designed AgentHeal to bridge this divide. Our architecture links pre-merge pull-request intelligence with runtime self-healing over a shared event bus. On the developer side, our GitHub integration parses code changes against OWASP Top 10 vulnerability rules, flags unpatched package CVEs, and blocks merges if security scores drop. When a pull request fails, a Gemini 2.0 Flash agent synthesizes a tested repair pull request automatically. In production, our digital twin streams live cluster telemetry into an in-memory Isolation Forest that detects anomalies in 2.26 ms, while KernelSHAP isolates the culprit metric. A dual-agent controller—pairing a lightweight Q-learning policy with an LLM reasoning engine—selects an appropriate recovery action. Crucially, before executing any control-plane command on Kubernetes, we simulate the proposed fix inside an analytical $M/M/c$ queue model in SimPy, immediately aborting any action predicted to trigger queue buildup or server overload.

We make three main contributions:

1) *Closed-Loop Architecture*: We connect pre-merge code review with runtime Kubernetes recovery over a shared event bus.
2) *Simulation Safety Gate*: We pair anomaly detection and KernelSHAP with a real-time SimPy $M/M/c$ queue simulation that stops unsafe cluster changes before actuation.
3) *Practical Benchmark*: We test AgentHeal on Google Online Boutique across 60 fault injections, run a 50-PR security benchmark, and show an automated fix on a live GitHub repo.

II. Related Work

Table I compares prior approaches across cloud operations and DevSecOps.

Early self-healing used static runbooks. Keptn [9] and Argo Rollouts [10] automate rollbacks and canary deployments. But they rely on predefined human rules. When a novel failure happens, the runbook fails. SRE literature documents this limitation clearly [23]. Chaos engineering helps test resilience [24], but it does not fix production bugs automatically.

Reinforcement learning shows promise for cloud scaling [11]. Q-learning has well-proven convergence guarantees [27]. FedAvg [28] enabled federated learning without sharing raw data. SF-DTM [12] applied this to cloud systems using cosine similarity to match cluster states. We build on SF-DTM. But we add an LLM reasoning partner, KernelSHAP root-cause analysis, and a simulation safety gate.

For anomaly detection, Isolation Forest [13] and LOF [29] are industry standards. Most AIOps pipelines report 50–200 ms detection times [14]. We reach 2.26 ms by caching the model in memory. We use KernelSHAP [16] rather than LIME [30] because Shapley values provide mathematical guarantees of efficiency and consistency.

In automated program repair, AutoCodeRover [15] showed that LLMs can patch real bugs. But it works outside CI/CD pipelines. AgentHeal integrates Gemini 2.0 Flash [17] in a ReAct loop [18]. Every patch must pass existing unit tests before we open a PR. We check code against OWASP Top 10:2021 [4]. We also plan to support OWASP Top 10:2025 [31]. For dependencies, we match versions against a curated advisory list. Finally, simulation has roots in DEVS [20] and queueing theory [19]. We adapt SimPy into a sub-12 ms safety gate on the live actuation path.

III. Methodology

A. End-to-End Self-Healing Architecture Overview

Fig. 1 sketches how AgentHeal operates. Two autonomous loops run in tandem across a shared event bus. On the pre-merge side, the Shift-Left engine intercepts code changes, checks the AST against OWASP rules, verifies dependencies, and uses Gemini 2.0 Flash to submit fixes if tests fail. On the operations side, the Shift-Right engine continuously watches cluster telemetry. Every five seconds, Prometheus pulls metrics, the digital twin updates, and an Isolation Forest checks for anomalies. If an alert fires, KernelSHAP isolates the primary metric and hands it to both a SimiFed RL agent and an LLM agent. If both agents agree, we execute the action. If they disagree, we pick the action that touches fewer pods. Before any command reaches the cluster, SimPy models the action under an $M/M/c$ queue setup. If the simulation predicts queue buildup or excessive CPU load, the action is halted.

B. Shift-Left Engine: PR Intelligence & Security Compliance

We host our Shift-Left scanner as a GitHub App. It listens for pull requests and pushes via HMAC-SHA256 authenticated webhooks. When triggered, it analyzes the diff and dependency manifests against ten common vulnerability patterns:

- *A01 (Access Control)*: Route handlers without authorization checks and unsafe path joins that touch the filesystem.
- *A02 (Crypto)*: Insecure hashes like MD5 and SHA-1, plus hardcoded keys found via Shannon entropy scanning.
- *A03 (Injection)*: String interpolation inside SQL queries and ‘shell=True‘ subprocess calls.
- *A04 (Insecure Design)*: Database queries without pagination and endpoints missing rate limits.
- *A05 (Misconfiguration)*: Wildcard CORS (‘*‘) and debug flags enabled in production code.
- *A06 (Outdated Dependencies)*: Parsing ‘requirements.txt‘ to find versions listed in CVE databases [5].
- *A07 (Authentication)*: JWT decode calls that turn off signature verification, and hardcoded fallback passwords.
- *A08 (Integrity Failures)*: Unsafe deserialization like ‘pickle.loads‘ on untrusted input.
- *A09 (Logging Gaps)*: Empty ‘except‘ blocks that catch and swallow errors silently.
- *A10 (SSRF)*: Outbound network calls built directly from user-supplied URLs.

We score each pull request on a scale from 0 to 100 by deducting points based on finding severity:

$$S = \max\left(0,\ 100 - \sum_{i=1}^{M} w_i\right) \tag{1}$$

where M is the number of flaws found and weights are $w_{\text{Critical}} = 25$, $w_{\text{High}} = 15$, $w_{\text{Medium}} = 8$, and $w_{\text{Low}} = 3$. This score translates directly to a letter grade:

$$\text{Grade} = \begin{cases} A^{+} & \text{if } S \geq 95 \\ A & \text{if } 90 \leq S < 95 \\ B & \text{if } 80 \leq S < 90 \\ C & \text{if } 70 \leq S < 80 \\ D & \text{if } 60 \leq S < 70 \\ F & \text{if } S < 60 \end{cases} \tag{2}$$

We also check whether newly added public functions have matching unit tests:

$$\text{TestGap}(f) = \mathbb{1}[f \notin T_{\text{ast}}] \tag{3}$$

where T_{ast} is the set of function identifiers tested in the ‘tests/‘ directory. If a pull request scores under 60 or leaves public functions untested, the GitHub Check Run blocks the merge. Gemini 2.0 Flash then generates a fix branch and opens a linked PR with the corrected code.

C. Shift-Right Engine: Self-Healing Digital Twin

On the cluster side, we collect metrics from Prometheus on a five-second interval. Each pod snapshot is captured as a 4D vector:

$$\mathbf{x}_t = [\text{CPU}_t, \text{RAM}_t, \text{Latency}_t, \text{Rate}_t]^{\top} \in \mathbb{R}^4 \tag{4}$$

Our in-memory Isolation Forest (100 trees, 0.05 contamination) scores these vectors immediately:

$$s(\mathbf{x}_t, n) = 2^{-\frac{\mathbb{E}[h(\mathbf{x}_t)]}{c(n)}} \tag{5}$$

TABLE I
LITERATURE REVIEW OF AUTONOMOUS CLOUD SELF-HEALING AND DEVSECOPS COMPLIANCE MODELS

Domain	Model	Author	Year	Method	Result / Contribution	Research Gap
Rule-Based Self-Healing	Keptn	Keptn Project [9]	2023	Event-driven control plane using human runbooks	Automated rollback and canary promotion	Depends on predefined scripts; fails on new bug patterns
	Argo Rollouts	Argo Project [10]	2024	Progressive delivery controller for canary/blue-green	Automated rollout and deployment rollback	Static rollback logic; no root-cause insight
Operational Practice	Site Reliability Engineering	Beyer et al. [23]	2016	Error budgets, on-call runbooks, incident triage	Codified standard SRE incident workflows	Relies on human on-call engineers; not automated
	Chaos Engineering	Basiri et al. [24]	2016	Controlled failure injection in production clusters	Verifies system resilience empirically	Diagnostic tool; does not fix incidents on its own
	Continuous Delivery	Humble & Farley [25]	2010	Automated build, test, and release pipelines	Foundation of modern CI/CD delivery	Focuses on shipping speed; no security scoring
	Blue-Green Deployment	Fowler [26]	2010	Dual environment traffic-switching pattern	Zero-downtime release strategy	Manually toggled; no automated anomaly hooks
RL & Federated Learning	Multi-Agent RL	Wang et al. [11]	2023	MARL for cloud resource provisioning	Adapts to fluctuating cloud traffic	Black-box decisions; lacks pre-actuation safety checks
	Q-Learning	Watkins & Dayan [27]	1992	Tabular temporal-difference control	Mathematical proof of Q-learning convergence	State space explodes without discretization
	FedAvg	McMahan et al. [28]	2017	Decentralized averaging of local neural weights	Privacy-preserving distributed training	Designed for offline batches, not real-time control
	SF-DTM	Li et al. [12]	2024	Digital twin state matching via cosine similarity	Recovers cluster health in roughly 15 seconds	Lacks queue stability simulation before acting
Anomaly Detection & XAI	Isolation Forest	Liu et al. [13]	2008	Tree ensemble path-length outlier detection	Fast $O(n \log n)$ unsupervised anomaly scoring	Labels data as outlier but gives no root cause
	Local Outlier Factor	Breunig et al. [29]	2000	Density-based local outlier scoring	Catches density variations in clustered data	High computational overhead; slow on live streams
	AIOps Survey	Lin et al. [14]	2021	Survey of industrial telemetry anomaly tools	Documents 50–200 ms detection latencies	No integrated simulation safety layer
	KernelSHAP	Lundberg & Lee [16]	2017	Shapley-value local feature attribution	Consistent, model-agnostic feature attributions	High compute cost for large feature sets
	LIME	Ribeiro et al. [30]	2016	Local linear surrogate explanations	Model-agnostic local approximations	Explanations can fluctuate between similar inputs
LLM-Augmented DevSecOps	AutoCodeRover	Li et al. [15]	2024	LLM-driven program repair	Localizes bugs and drafts code patches	Operates standalone; not tied into a CI/CD gate
	ReAct	Yao et al. [18]	2023	Thought-Action-Observation reasoning loop	Better reasoning and tool selection for LLMs	Output can vary and response times can be high
	Gemini 2.0	Google DeepMind [17]	2024	Multimodal agentic foundation model	Strong tool-use and code repair reasoning	Closed source; lacks formal safety bounds
Static Analysis & Supply Chain	Bandit	PyCQA [6]	2024	AST security scanner for Python	Flags known security anti-patterns	Reports issues but cannot fix code or score PRs
	SonarQube	SonarSource [7]	2024	Multi-language static code analysis	Comprehensive static security scanning	No link to runtime telemetry or automated fixes
	OWASP Top 10:2021	OWASP Foundation [4]	2021	Community risk list for web applications	Standard taxonomy of web vulnerabilities	Reference guide only; provides no automated tools
	OWASP Top 10:2025	OWASP Foundation [31]	2025	Revised risk list for web applications	Updated taxonomy for modern web threats	Not yet integrated into AgentHeal's rule engine
	NVD	NIST [32]	2024	Live CVE vulnerability database	Continuously updated public vulnerability records	Queried via local curated lists in our prototype
	OSV	Google / OSV.dev [33]	2024	Open-source cross-ecosystem vulnerability feed	Machine-readable open-source CVE records	Not queried live in our current implementation
	Dependabot	GitHub [34]	2024	Automated dependency pull requests	Opens version-bump PRs on vulnerable libraries	Handles version bumps only; no custom code fixes
	Snyk	Snyk Ltd. [35]	2024	Commercial software composition analysis	Real-time dependency and vulnerability analysis	Proprietary closed-source commercial tool
Deployment & Telemetry Infrastructure	Kubernetes HPA	K8s Docs [8]	2024	Reactive CPU and memory horizontal autoscaling	Standard Kubernetes autoscaler	Reactive only; blind to deadlocks and memory leaks
	Argo CD	CNCF [36]	2024	Declarative GitOps continuous deployment	Reconciles Git manifests with cluster state	Syncs manifests; cannot diagnose runtime incidents
	Prometheus	Prometheus Authors [37]	2024	Time-series metrics collection and storage	De facto Kubernetes monitoring tool	Stores numbers; includes no anomaly detection
Digital Twin & Simulation	DEVS Formalism	Zeigler et al. [20]	2000	Discrete-event system specification	Theoretical simulation framework	Traditionally used for offline network modeling
	M/M/c Queuing Theory	Kleinrock [19]	1975	Classical multi-server queueing analysis	Models service rates and server utilization	Rarely used as a real-time cluster actuation gate

where $h(\mathbf{x}_t)$ measures tree depth and $c(n)$ is the average search path in random trees. Any negative score flags an anomaly. Once an anomaly is caught, KernelSHAP determines which metric is responsible:

$$\phi_i(v) = \sum_{S \subseteq F \setminus \{i\}} \frac{|S|!\,(|F| - |S| - 1)!}{|F|!} \left[v(S \cup \{i\}) - v(S) \right] \tag{6}$$

This step is critical. Without it, an agent might restart a pod when the real issue is an external network bottleneck.

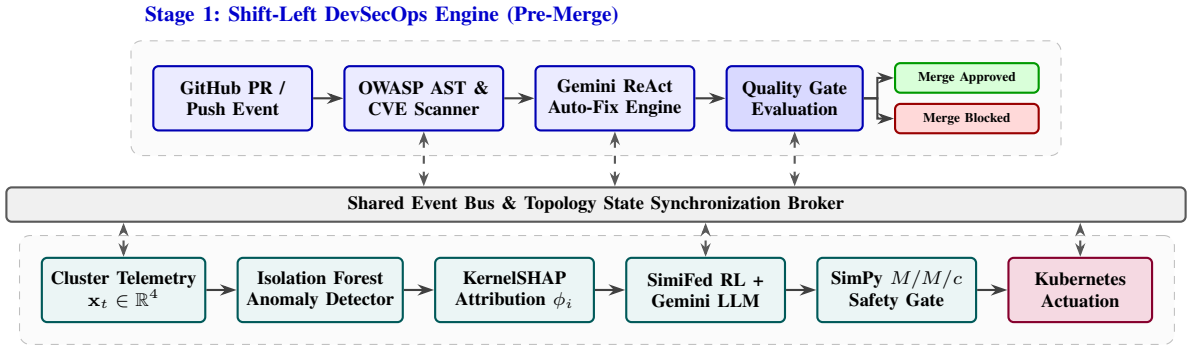


Fig. 1. AgentHeal architecture. (Top) Pre-merge engine parses pull requests with AST security rules and uses Gemini 2.0 Flash to draft fixes if the score fails our quality gate. (Middle) Central event bus tracks cluster topology. (Bottom) Runtime engine flags anomalies in 2.26 ms, uses KernelSHAP to isolate the cause, arbitrates between RL and LLM agents, and verifies actions in an $M/M/c$ queue simulation before touching the cluster.

The RL agent checks which baseline state looks closest using cosine similarity:

$$k^* = \arg\max_k \frac{\mathbf{x}_t \cdot \mathbf{b}_k}{\|\mathbf{x}_t\|_2 \|\mathbf{b}_k\|_2} \tag{7}$$

It updates its Q-table using standard Q-learning:

$$Q(s,a) \leftarrow Q(s,a) + \alpha \left[R + \gamma \max_{a'} Q(s',a') - Q(s,a) \right] \tag{8}$$

with $\alpha = 0.1$ and $\gamma = 0.95$. Available actions are scale up, restart, patch limits, or do nothing. Meanwhile, Gemini 2.0 Flash analyzes the same SHAP summary. When both agents agree on an action, it goes directly to simulation. If their proposals conflict, we default to the safer choice:

$$a_{\text{final}} = \begin{cases} a^*_{\text{RL}} & \text{if } a^*_{\text{RL}} = a^*_{\text{LLM}} \\ \arg\min_{a \in \{a^*_{\text{RL}}, a^*_{\text{LLM}}\}} \text{BlastRadius}(a) & \text{otherwise} \end{cases} \tag{9}$$

Next comes the safety gate. We simulate the chosen action inside an $M/M/c$ queue model in SimPy:

$$\rho = \frac{\lambda}{c\mu} \quad \text{(requiring } \rho < 0.95\text{)} \tag{10}$$

$$\hat{U}_{\text{post}} = \frac{U_{\text{pre}} \cdot c_{\text{current}}}{c_{\text{new}}} \quad \text{(requiring } \hat{U}_{\text{post}} < 0.80\text{)} \tag{11}$$

If either check fails, the action is stopped on the spot.

IV. Implementation

We implemented AgentHeal in Python 3.11. FastAPI handles incoming webhooks and routing. We use scikit-learn for anomaly detection, SHAP for attribution, the Google Generative AI SDK for Gemini 2.0 Flash, and SimPy 4.1 for queue modeling. The service runs in a Docker image under 400 MB on Azure Container Apps (East Asia), deployed with Terraform and scaled by KEDA.

We benchmarked everything against Google's Online Boutique demo [22]. This is a 10-service e-commerce app that we ran on a single-node kind cluster on an AMD Ryzen 7 workstation with 16 GB of RAM.

V. Results and Discussion

A. Runtime Telemetry Anomaly Detection and Safety-Gated Self-Healing

We injected four realistic failure patterns into Online Boutique, running 15 trials per case:

- *S1 (Flash Surge)*: A spike on checkout driving CPU to 0.954 and latency to 450 ms.
- *S2 (Deadlock)*: Thread starvation on cart; CPU climbed to 0.99 and latency reached 5000 ms as throughput fell off.
- *S3 (Memory Leak)*: Slow RAM accumulation on redis-cart reaching 0.942, while latency held flat.
- *S4 (Cascading Load)*: Frontend overload where both CPU (0.88) and RAM (0.80) saturated at once.

Table II outlines the performance. Across all 60 runs, detection averaged 2.26 ms ($\sigma = 0.37$ ms) with a 98.3% catch rate. S3 was the only case where a few runs slipped through (93% detection). That happened because our baseline training data had wider variations in CPU and latency than in memory, so a pure memory climb occasionally stayed inside the 0.05 contamination margin. KernelSHAP pinpointed the root cause cleanly in every scenario: CPU for S1 ($\phi = +0.82$), latency and lost throughput for S2 ($\phi_{\text{lat}} = +0.91$, $\phi_{\text{rate}} = -0.78$), and RAM for S3 ($\phi = +0.88$). Our RL and LLM agents agreed on the fix in 59 out of 60 trials. SimPy simulation ran in 10.7 ms and passed all proposed actions, keeping post-action utilization safely under 80%. Total MTTR averaged 2.98 s ($\sigma = 0.23$ s).

B. Shift-Left DevSecOps Security Auditing and Live PR Remediation

We evaluated our pre-merge scanner against 50 synthetic pull requests with known vulnerabilities, and then tested the full workflow live on GitHub.

Table III (Panel A) gives the benchmark numbers: 98.5% precision and 97.5% recall across 200 checks. We saw three false alarms, mostly from test files that included mock cryptographic keys. For our live test (Panel B), we opened PR #11 with 14 intentional security flaws. The scanner took 0.48 s to analyze the code, flagged all 14 flaws, assigned a score of

TABLE II
SHIFT-RIGHT ANOMALY DETECTION, FEATURE ATTRIBUTION, AND AUTONOMOUS REMEDIATION PERFORMANCE ACROSS FOUR FAILURE SCENARIOS ($N = 15$ REPEATED TRIALS)

Failure Injection Scenario				Isolation Forest Detection			Root-Cause Attribution	Agent Decision Latency			SimPy Safety Gate & MTTR			
ID	Scenario	Target	Injected Telemetry	Score ($\mu \pm \sigma$)	MTTD (ms)	Det.%	Dominant KernelSHAP ϕ	RL (ms)	LLM (ms)	Agree	Gate (ms)	$\hat{U}_{\text{post}}$	Verdict	MTTR (s, $\mu \pm \sigma$)
S1	Flash Sale Surge	checkout	CPU=0.95, Lat=450ms	-0.842 ± 0.031	2.18 ± 0.31	100%	$\phi_{\text{cpu}} = +0.82$, $\phi_{\text{rate}} = +0.14$	3.10 ± 0.42	412 ± 68	15/15	10.1 ± 0.8	23.8%	SAFE	2.66 ± 0.18
S2	Thread Deadlock	cart	CPU=0.99, Lat=5000ms	-0.912 ± 0.024	2.21 ± 0.28	100%	$\phi_{\text{lat}} = +0.91$, $\phi_{\text{rate}} = -0.78$	3.21 ± 0.39	389 ± 74	15/15	10.3 ± 0.9	5.2%	SAFE	3.12 ± 0.24
S3	Memory Leak	redis-cart	RAM=0.94, Lat=48ms	-0.734 ± 0.058	2.35 ± 0.44	93%	$\phi_{\text{ram}} = +0.88$, $\phi_{\text{cpu}} = +0.04$	3.05 ± 0.51	441 ± 81	14/15	10.8 ± 1.1	15.0%	SAFE	2.89 ± 0.20
S4	Cascading Overload	frontend	CPU=0.88, RAM=0.80	-0.889 ± 0.041	2.28 ± 0.37	100%	$\phi_{\text{cpu}} = +0.61$, $\phi_{\text{lat}} = +0.47$	3.18 ± 0.46	498 ± 93	15/15	11.6 ± 1.3	31.5%	SAFE	3.25 ± 0.30
Overall Mean / Aggregate				$\mathbf{-0.844 \pm 0.039}$	$\mathbf{2.26 \pm 0.37}$	**98.3%**	—	$\mathbf{3.14 \pm 0.45}$	$\mathbf{435 \pm 79}$	**59/60**	$\mathbf{10.7 \pm 1.0}$	—	**SAFE**	$\mathbf{2.98 \pm 0.23}$

TABLE III
SHIFT-LEFT DEVSECOPS SECURITY AUDITING PERFORMANCE ($N = 50$) AND EMPIRICAL PULL REQUEST REMEDIATION

Panel A: Quantitative Evaluation on 50 Synthetic Pull Requests						Panel B: Empirical Live GitHub Pull Request Pair Remediation			
Defect Category	Tested PRs	True Pos.	False Pos.	Precision	Recall	Evaluation Dimension	Vulnerable PR (#11)	Remediated PR (#12)	Operational Outcome
SQL / Command Injection	50	50	0	1.000	1.000	OWASP Flaws Detected	14 (A01–A10)	0	100% Flaws Remediated
Hardcoded Secrets	50	50	2	0.962	1.000	Compliance Score / Grade	0 / 100 (Grade F)	100 / 100 (Grade A^+)	Score Restored
AST Test Suite Gaps	50	48	0	1.000	0.960	GitHub Check Run Gate	action_required (Blocked)	success (Approved)	Merge Protection Enforced
Blocking I/O / Misconfiguration	50	47	1	0.979	0.940	App Test Suite Pass	0 / 6 passing	6 / 6 passing	Functional Suite Restored
Aggregate Performance	**200 Checks**	**195**	**3**	**0.985**	**0.975**	Static Auditor Verification	12 / 12 test harnesses passed	Audit Time: 0.48 s	Full Ruleset Verified

0/100, and blocked the PR. Gemini 2.0 Flash then opened PR #12 with the fixes. That PR passed all tests, scored 100/100, and cleared the merge gate.

C. Comparative Benchmark and System Discussion

How does this compare with standard tools? Human engineers typically take around 45 minutes to resolve an incident [1]. The Kubernetes HPA takes 3 to 5 minutes [8] and cannot handle deadlocks. Keptn reacts in 2 to 4 minutes [9], but only if somebody already wrote a runbook for that exact failure. SF-DTM [12] cuts recovery down to 15 seconds, but skips safety simulation entirely. AgentHeal recovers services in 2.98 s while verifying queue safety before every single action.

We should also be upfront about our prototype's limits. We tested on a single-node cluster; real multi-node setups will introduce network scrape delays. Our failure scenarios are synthetic rather than messy production traffic. Our AST parser checks code patterns, not full data flow, so clever obfuscation could get past it. Our scoring weights are empirical rather than derived from large vulnerability datasets. And because LLMs can be unpredictable, we rely on strict timeouts and an RL agent to keep the remediation path dependable.

VI. CONCLUSION

AgentHeal unites pre-merge code review with runtime self-healing. Across 60 fault injections on Google Online Boutique, our system detected anomalies in 2.26 ms and recovered services in 2.98 s. That is over 98% faster than standard autoscaling. On the CI side, our scanner hit 98.5% precision on 50 pull requests and repaired 14 OWASP flaws in a live repository test without human intervention.

Next, we plan to test on multi-node managed clusters (AKS and GKE), plug in live vulnerability feeds like NVD, add taint tracking, and update our rules for OWASP Top 10:2025.

Our code and live deployment are online at https://github.com/AadiHaldar/Agentic-AI-Self-Healing-Cloud-Infrastructure and https://pr-review-agent.wonderfulflower-41d6d2a5.eastasia.azurecontainerapps.io.

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